

I. Introduction and Background

Wine is a delicious alcoholic beverage which has been consumed by humans for millennia. Produced in many regions around the world including California and Europe, wine offers a rich variety of flavors and characteristics. Wine quality has traditionally been judged by experts who assess its taste characteristics. While these expert opinions are valuable, there is a growing interest in understanding the measurable chemical properties that influence wine quality.

Our project focuses on the analysis of wines from the Northern Region of Portugal called Vinho Verde. According to master class, “Vinho Verde (pronounced “veeng-yo vaird-ay”) is a Portuguese wine that comes from the region of Vinho Verde, a Denominação de Origem Controlada (DOC), which is the country’s largest appellation.” These wines come in the form of reds, whites, or roses.

White Vinho Verde wines conjure up refreshing tastes. The tasting notes include “Lemonade, White Melon, Gooseberry, Grapefruit, and Lime Blossom” (Puckette). The Beginner’s Guide To Vinho Verde Wines also states that “six grapes (you’ve never heard of) dominate the regional blend: Alvarinho, Arinto, Azal, Avesso, Loureiro, and Trajadura.” Even though the data set we are analyzing does not discuss grape variety, grape variety influences the wine’s flavor.

In addition to Vinho Verde wines being a fun summer wine to drink, their fermentation process is also fascinating. The author further explains, “Most Vinho Verde today is artificially carbonated. Originally though, the carbonation happened when winemakers bottled so quickly that the fermentation would finish in the bottle. Today, you will find higher end producers shying away from carbonation all together” (Puckette).

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By using statistical analysis, we can explore how these objective measurements relate to subjective quality rating. Our study examines Vinho Verde Wines in the Portugal region using data from the UC Irvine Machine Learning Repository. We look at both red and white varieties to understand how their chemical properties relate to wine quality ratings. By analyzing this dataset, we aim to determine which chemical properties best predict wine quality and whether these factors differ between red and white varieties.

We use several statistical methods in our analysis. First, we explore the data to understand the variables, their types and their relationships. Then, we use multivariate regression to identify which factors significantly predict wine quality. We also compare red and white varieties to find meaningful differences that contribute to their quality.

The other way we will compare variables for red and white wines in this data set is through analysis of variance or (ANOVA). The results of our analysis have practical implications. For winemakers, understanding which chemical properties influence wine quality can help them make production decisions to better serve their markets. For consumers, this analysis adds to our understanding of the science behind wine enjoyment.

This report presents our complete analysis, starting with a description of the data, followed by our data analysis and conclusions. Our goal is to use statistics to better understand what makes a good wine.

II. Description of the data

Our analysis uses the wine quality data set from the UC Irvine Machine Learning Repository. The dataset consists of two separate files: one for red wine with 1,599 observations and the other for white wine with 4,898 observations. The dataset also includes 11 chemical properties along with a wine quality rating. The quality ratings range from 0 (very poor) to 10 (excellent).

For our analysis, we combined the red and white datasets to include a categorical variable, color. This allows us to examine whether the chemical properties and quality of wine differ between red and white varieties. In total, there are 6,479 observations in our dataset.

Table 1 presents summary statistics for selected variables:

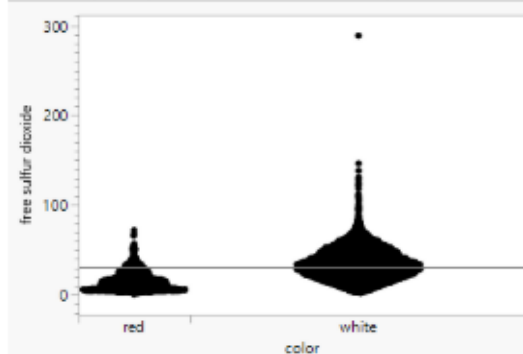
Table 1 Summary Statistics:

Variable	Mean	Median	SD	Min	Max
Quality	5.8	6	0.87	3	9
pH	3.2	3.21	0.16	2.72	4.01
Alcohol (%)	10.5	10.3	1.19	8	14.9
Residual Sugar	5.4	3	4.76	0.6	65.8

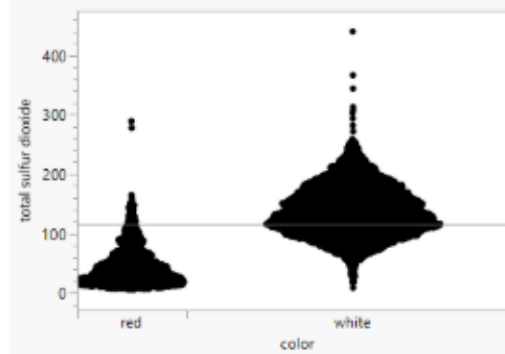
The quality variable shows a somewhat normal distribution. Although the range of possible values is 0-10, our observed data set had ratings ranging from 3-9 with most quality ratings falling in the 5-6 range.

III. Data Analysis

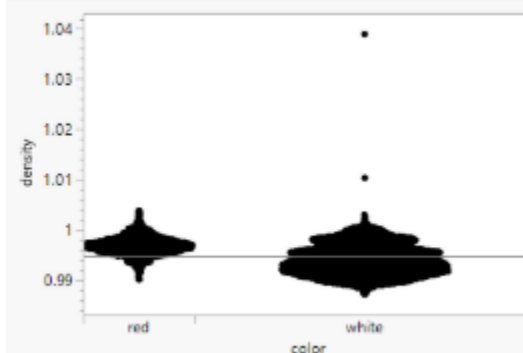
▼ Oneway Analysis of free sulfur dioxide By color



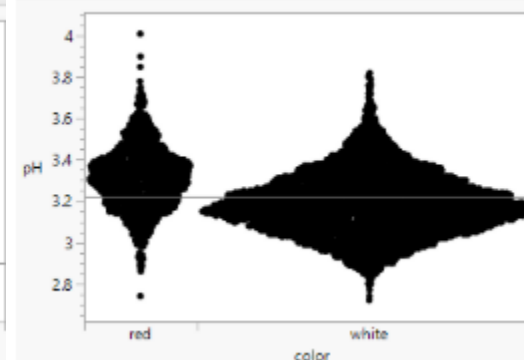
▼ Oneway Analysis of total sulfur dioxide By color



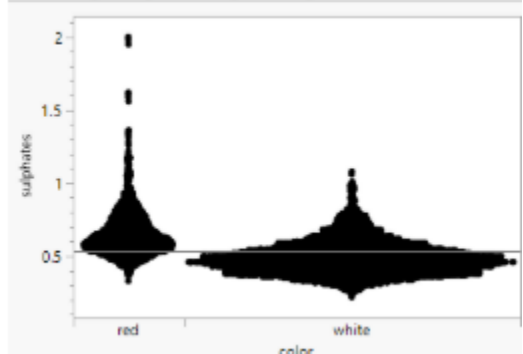
▼ Oneway Analysis of density By color



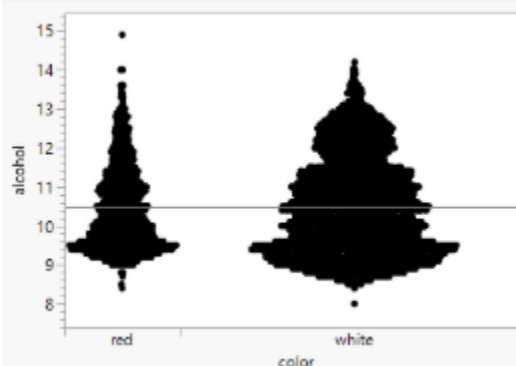
▼ Oneway Analysis of pH By color



▼ Oneway Analysis of sulphates By color



▼ Oneway Analysis of alcohol By color



▼ Oneway Analysis of quality By color



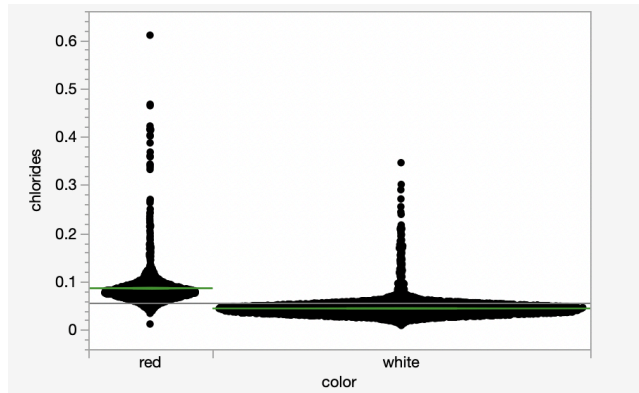
While fitting the ANOVA models in JMP above, the variables we used were free sulfur dioxide, total sulfur dioxide, density, pH, sulfates, alcohol, and quality and made them by colors red and white. After fitting all the models for these variables, it shows that white wines have higher means except for the standard errors. We can also tell that the models with respect to white wines have bigger shapes. Looking at the wine quality, white wines seem to have better quality than red wines due to their sweetness and taste. While most of the models are shaded in solid black, the quality variable has a graph using some lines. The model based on quality has the biggest difference due to how it is plotted. After making the models, we also test the hypothesis where the null hypothesis is that the relationship between red and white wines are the same for all the variables while the alternative hypothesis is that the relationship between red and white wines is different for all the variables. We also found out that our p-values are less than .0001 as well as having the degrees of freedom at 1 and 6495 for color and error respectively. And there are additional variables that were used as well.

Table 2 shows the Sum of Squares for sources on each respective variable:

Table 2: Sum of Squares

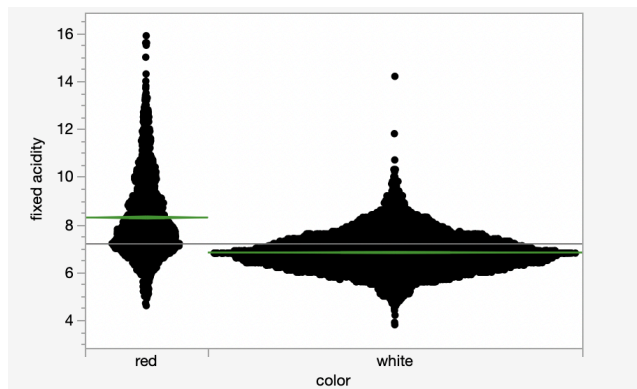
variables	color	error	total
Free sulfur dioxide	455241	1591266.6	2046507.6
Total sulfur dioxide	10179301	10573600	20752901
density	0.00891394	0.04949836	0.05841229
pH	18.192	149.746	167.938
sulfates	34.14544	109.69671	143.84215
alcohol	10.0448	9230.9135	9240.9583
quality	70.5308	4883.1549	4953.6857

Looking at the table, it shows that the density has the smallest sum of squares while the sulfur dioxides have the larger sums of squares overall.



The image on the left is an ANOVA analysis of fixed acidity for the wines, separated by color. We can see that the red wine is more variable, with a larger variation in fixed acidity, while white wine is more concentrated around an acidity of

7. The sum of squares for the wine color was 2586.667 with an error sum of squares of 8331.424, meaning the variability between and within groups was very large. Finally, our F-statistic produced a p-value of .0001, meaning there is evidence to suggest that the means for these two groups of wine color are in fact different.



We did another ANOVA analysis of chlorides with respect to wine color, and the results are shown on the graph on the left. Again, the red wine was more variable, while the white wine was more concentrated on .05. Our color sum of squares was 2.0956, and error

sum of squares was 5.877, which means the variability within the groups is larger than the variability between groups. Our F-statistic produced a p-value of less than .0001, which means we reject the null hypothesis and conclude that the means are different for average chlorides for the two colors of wine.

This trend we see in our ANOVA analyses continues on for each variable we examined. The red wines tended to have larger variability within themselves, often appearing stretched vertically, while the white wines were often more concentrated around a specific value, stretching their graph horizontally. We arrived at the same conclusion for each ANOVA analysis, which is to reject our null hypothesis that the means for the different colors of wine were the same for each variable. With consistent p-values of less than .0001, we are confident that the means are not the same for each color.

Finally, in our data analysis, we conducted both a forward and backward selection process to find the best fit model for our data. We wanted a model that used pH as a response variable and most effectively modeled the data. When we did backward selection, we ended up with the original full model, as no variable had a p-value that was above the “p-to-leave” threshold. To confirm this, we did a forward selection as well and ended up with the same results, with all of our predictor variables to the pH response variable.

Before performing a backwards selection algorithm on the linear regression model with all 12 variables, we made a simple linear regression model. The model in question had free sulfur dioxide as the response and total sulfur dioxide as the predictor. Our assumption for this was that we assumed there would be a total amount of sulfur dioxide in the wine. Consequently there would be some of that sulfur dioxide not fully absorbed in the wine. Naturally, the free sulfur dioxide would be proportionate to the total sulfur dioxide. After constructing the model in R, my R-squared was 0.5197. Thus 51.97% or approximately 52% of the variation in this model can be explained by the relationship of free sulfur dioxide to total sulfur dioxide.

Analyzing categorical variables in the context of statistics and data science is not easy. Two ways that come to mind are to make the categorical variables in question into dummy

variables. Those categorical variables are assigned 1 or 0 depending on if the condition for the categorical variable is met. For example, in our class's lecture notes, attending a private high school was 0 and attending a public high school was a 1. The linear F-test is important because this test allows statisticians to determine which variables are important and which can be eliminated.

The F-test is conducted as follows. According to lesson 6.2- The General Linear F Test, there is importance in first defining a larger full model. Furthermore, a larger model has more parameters than the reduced model. Next define a model with fewer parameters than the large one. (Lesson 6.2) After that, conduct the F-test. In order to perform the F-test statistic do the following:

First, "Fit the full model" to the data." (Lesson 6.2)

Obtain the least squares estimates of X and X_0 . "Determine the error sum of squares, which we denote as "SSE(F)." (6.2)

After that, "Fit the reduced model" to the data." (Lesson 6.2)

The last quantity to find would be SSE. (Lesson 6.2) Most importantly, we reject the null hypothesis that the reduced model is better than the full model if F is large. Also if the corresponding p-value is small, we rFor our project I performed a background selection algorithm on a model that held all 11 variables that corresponded to the pH. My variables were fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, sulphates, alcohol, color, and quality respectively. Since I was working in R, I fit the respective model, found the residuals, formulated the SSE and finally obtained my F

value. The F value was obtained through the summary function in R. Additionally my null hypothesis for each model was that the reduced model was better than the original model at explaining the variation between the responses and their respective predictors. The F statistics for each respective model are as follows, 851.4, 921.9, 299.6, 558 336.4 , 317 339.6, 403.8, 642 486.3, and 443. These results show that the F-statistics are extremely large. The original model is consequently the most accurate model for our purposes. Last but not least, the original model has the highest R^2 and adjusted R^2 values.

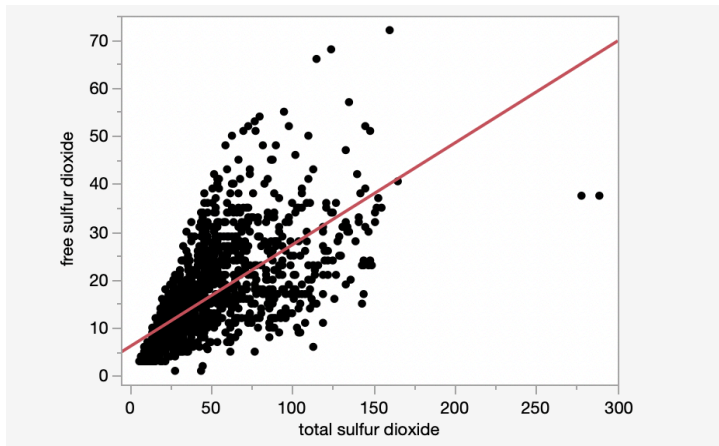
The model with the highest R and adjusted R squared values would be the model that can explain the most variation between the response and the predictors.

IV. Conclusion

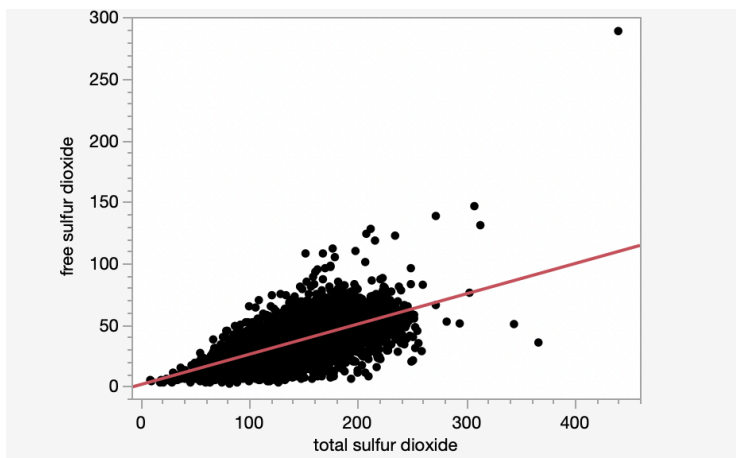
Our group still has further research questions related to the physical and chemical properties of Vinho Verde wines.

The main topics of inquiry would be grape type and soil type. These subjects were not elaborated on by our project. Our first question would be does grape type have an effect on pH for these wines? Additionally if pH were dependent on grape type would such an effect differ based on color? Similar questions could be asked regarding citric acid and chlorides. The most important question regarding grape type is as follows. How much of the variation in wine quality can be explained by the type of grape alone? Finally, does the average amount of sulphates in the grapes vary by color? While investigating the regression model, we have found that red wines on average appear to have fewer sulfates than white wines.

The type of soil used to grow Vinho Verde wines is important.



(red) simple linear regression for
 free sulfur dioxide = $6.0095 + .2123$
 $\times$ total sulfur dioxide
 $R^2 = .4458$



(white) simple linear regression for
 free sulfur dioxide = $1.2278 +$
 $.2463 \times$ total sulfur dioxide
 $R^2 = .3787$

For instance, does elevation play a role in the production of either red or white wines?
 Furthermore, is there a difference in the average amount of sulphates in the soil that can be
 explained by an ANOVA model? Additionally, where would the chlorides come from? Finally,
 how much variation in wine quality can be explained by soil alone?

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